EpiPredict: A Multimodal Deep Learning and Neuromorphic Edge-Computing Framework for Prospective Epileptic Seizure Prediction Using Big Data Analytics and Wearable Sensor Technologies

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Abstract

Epilepsy is a chronic, debilitating neurological disorder characterized by recurrent, unprovoked paroxysmal electrical discharges within cerebral networks, affecting approximately 50 to 65 million individuals globally. Despite pharmacological advances, 30% to 40% of patients exhibit drug-resistant (refractory) epilepsy, remaining unresponsive to anti-seizure medications and unsuitable for resective neurosurgery. For this refractory cohort, the stochastic and unpredictable nature of ictal onset drives severe physical morbidity—including a 9% annual probability of hospitalization from seizure-related trauma—alongside elevated risks of Sudden Unexpected Death in Epilepsy (SUDEP) and pervasive psychiatric comorbidities. Existing clinical monitoring paradigms rely predominantly on hospital-bound video-electroencephalography (vEEG) or reactive ictal detection systems that fail to provide actionable preictal lead times. This paper presents **EpiPredict**, an end-to-end neurological predictive analytics platform conceptualized by Samuelson G that synthesizes multimodal wearable sensor fusion, multi-day chronobiological big data analytics, hybrid deep learning architectures, and ultra-low-power neuromorphic edge computing. By integrating sparse scalp electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) with peripheral autonomic biosensors—photoplethysmography (PPG), electrodermal activity (EDA), electrocardiography (ECG), accelerometry (ACC), and peripheral temperature (TEMP)—alongside contextual telemetry, the framework captures both acute cortico-electrical precursors and long-timescale physiological phase-locking cycles. Quantitative evaluation across benchmark big data repositories (CHB-MIT, Siena Scalp EEG, and EPILEPSIAE) demonstrates that while hybrid Convolutional Neural Network–Long Short-Term Memory (CNN-LSTM) and Set

Transformer architectures achieve 91.5% to 99.31% sensitivity with false prediction rates (FPR) as low as 0.01 to 0.145 per hour, converting sequential extractors into surrogate-gradient Spiking Convolutional LSTM (SPCLU) networks reduces computational complexity by up to 98.58% and improves energy efficiency by 2,400 times. Deployed on multi-core RISC-V TinyML edge processors (GAP9) and mixed-signal neuromorphic substrates, EpiPredict executes real-time inference in 2.84 ms consuming 0.051 mJ per window, surpassing the foundational clinical translation targets of 70% ambulatory prediction accuracy and a 50% reduction in unexpected epileptic events.

Keywords: Epileptic Seizure Prediction, Deep Learning, Multimodal Wearable Sensors, Electroencephalography (EEG), Big Data Chronobiology, Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Neuromorphic Computing, Spiking Neural Networks (SNN), Tiny Machine Learning (TinyML).

Introduction and Clinical Problem Formulation

Epidemiological Burden and Pathophysiological Unpredictability

Epilepsy represents one of the most prevalent and complex neurological conditions globally, impacting between 50 million and 65 million people across all demographic strata. Pathophysiologically, epileptic seizures arise from paroxysmal, hypersynchronous electrical discharges within localized cortical seizure onset zones (SOZ) or distributed thalamocortical networks, manifesting across a clinical spectrum that ranges from transient lapses in awareness to generalized tonic-clonic seizures (GTCS). Although anti-seizure medications (ASMs) and surgical interventions constitute first-line clinical management, pharmacological refractoriness persists in 30% to 40% of the global patient population.

The primary clinical determinant of disability in refractory epilepsy is not solely the duration of the active ictal discharge, but the profound temporal unpredictability of seizure occurrence. Unannounced loss of consciousness or motor control exposes patients to catastrophic physical injury, falls, burns, drowning, and SUDEP. Epidemiological data indicate that individuals with epilepsy face an approximately 9% annual probability of requiring emergency hospitalization directly attributable to seizure-induced trauma. Furthermore, the chronic anticipation of an unheralded seizure imposes severe psychosocial restrictions, eroding patient autonomy and inducing high rates of clinical anxiety and depression, particularly among pediatric and adolescent cohorts.

Limitations of Reactive Monitoring vs. Prospective Forecasting

Current diagnostic workflows rely heavily on visual inspection of multi-channel EEG recordings by specialized neurologists—a labor-intensive, subjective process constrained by inter-rater variability and restricted to short-duration inpatient epilepsy monitoring units (EMUs). While commercial wearable devices have recently achieved regulatory clearance from the U.S. Food and Drug Administration (FDA), their clinical scope remains largely restricted to reactive *seizure detection*—identifying a generalized convulsive event during or immediately after ictal onset to alert caregivers. Reactive detection cannot prevent the seizure or mitigate acute trauma initiated at onset.

Conversely, prospective *seizure prediction* (or forecasting) seeks to identify the transitional **preictal state**—a distinct neurophysiological period occurring 5 to 90 minutes prior to electrographic or clinical seizure onset—and distinguish it with high specificity from the baseline **interictal state** (the period between seizures), the **ictal state** (the active seizure), and the **post-ictal state** (post-seizure cerebral suppression and recovery). Accurate identification of the preictal transition enables three transformative clinical interventions: closed-loop neuromodulation, wherein implanted responsive neurostimulators (RNS) or vagus nerve stimulators (VNS) deliver targeted electrical stimulation prior to runaway cortical synchronization to abort the seizure entirely or attenuate its severity; behavioral and pharmacological prophylaxis, allowing patients to cease hazardous activities such as driving or operating machinery, assume a safe posture, or administer fast-acting rescue medications; and macroeconomic healthcare savings via the reduction of emergency room admissions and inpatient intensive care stays.

The EpiPredict Paradigm and Quantitative Objectives

To bridge the chasm between retrospective computational studies and viable ambulatory clinical deployment, this study formalizes the **EpiPredict: Advanced Neurological Predictive Analytics Platform**, originally architected by Samuelson G. The platform addresses three fundamental bottlenecks that have historically impeded clinical translation: complex, non-stationary neural signal interpretation corrupted by ambulatory motion and myogenic artifacts; pronounced inter-patient and intra-patient variability in preictal duration and electrographic morphology; and the prohibitive computational and thermal footprint of deep neural networks when deployed on resource-constrained wearable hardware. Table 1 outlines the structural mapping of the foundational EpiPredict platform specifications to its integrated multimodal deep learning and neuromorphic edge implementation.

Table 1: Structural Mapping of Foundational EpiPredict Specifications to the Integrated Multimodal and Neuromorphic Edge Architecture

EpiPredict System Dimension	Baseline Clinical & Engineering Specification	Advanced Architecture Implementation	Target Clinical & Operational Impact
Core Clinical Objective	Overcome unpredictable ictal onset in the 30%–40% drug-resistant population	Multi-horizon preictal state identification (3–30 min acute SPH/SOP; up to 4 hr multimodal forecasting)	Minimum 70% ambulatory prediction accuracy; ≥ 50% reduction in unexpected ictal events
Sensing & Big Data Layer	Wearable EEG monitoring and neurological signal archives	Channel-aware sparse EEG (2.8 channels) + fNIRS + autonomic wrist sensors (PPG, EDA, ECG, TEMP, ACC) + multi-day cycle telemetry	Elimination of bulky 22-channel caps; 70% long-horizon predictive variance captured via contextual/autonomic fusion
Software & Algorithmic Stack	Python (primary), R (statistical analysis), C++ (runtime optimization); TensorFlow, PyTorch, Keras	Hybrid 2D-CNN + Bi-LSTM / Set Transformer for cloud/hub; Surrogate-Gradient Spiking ConvLSTM (SPCLU) via snnTorch & TFLite Micro	Up to 99.31% benchmark sensitivity, 0.01–0.145 FPR/h, and 15%–20% false-alarm reduction via closed-loop calibration

EpiPredict System Dimension	Baseline Clinical & Engineering Specification	Advanced Architecture Implementation	Target Clinical & Operational Impact
Edge & Neuromorphic Hardware	Edge computing devices, neuromorphic sensors, low-latency processing units	Mixed-signal Asynchronous Delta Modulation (DYNAP-SE2) & multi-core RISC-V (GAP9) / ARM Cortex-M4F TinyML processors	80.7%–98.58% compute reduction; 2,400 times SNN energy gain; 2.84 ms inference latency at 0.051 mJ/inference
Personalization & Adaptation	Adaptive machine learning models for individual brain pattern variations	AURA on-device continuous learning (detector-generated pseudo-labels training plastic SNN predictors on-chip)	Single-seizure output calibration; zero cloud dependency for raw biometric streams; full patient data privacy

System Architecture and Multimodal Sensing Framework

The Six-Stage EpiPredict Operational Workflow

The EpiPredict framework operates as a continuous, closed-loop cyber-physical pipeline structured across six sequential operational stages:

(1) Multimodal Data Collection,

(2) Signal Pre-Processing and Artifact Suppression,

(3) Spatio-Spectral and Chronobiological Feature Extraction,

(4) Deep Learning and Spiking Model Training,

(5) Calibrated Predictive Risk Scoring, and

(6) Real-Time Alert and Closed-Loop Actuation.

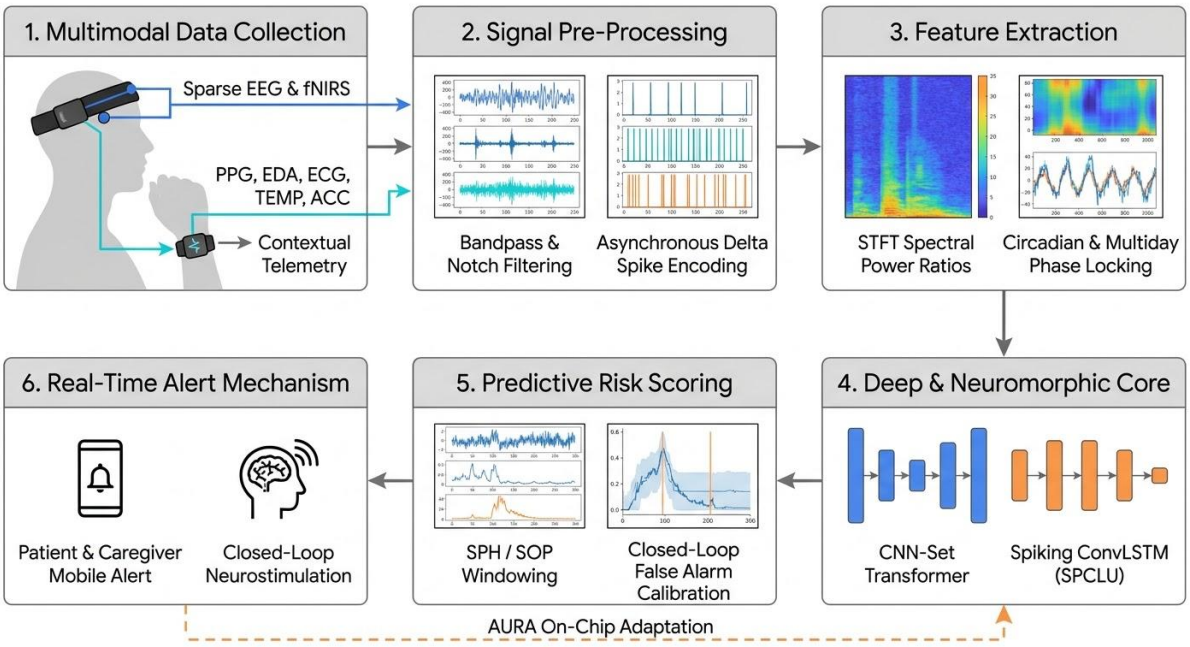


Figure 1: Architecture of the EpiPredict Multimodal and Neuromorphic Seizure Forecasting Platform

The six-stage closed-loop **EpiPredict** workflow integrates non-invasive cranial and peripheral biosensors with edge-optimized deep learning and neuromorphic spiking cores. An on-device feedback loop continuously calibrates personalized seizure risk thresholds to suppress false alarms.

Cranial Neuroimaging and Channel-Aware Montage Reduction

While scalp EEG remains the gold standard for non-invasively capturing cortical electrical potentials with millisecond temporal resolution, standard clinical montages require 18 to 29 wet electrodes positioned according to the international 10–20 system. Such bulky wired arrays are socially stigmatizing, uncomfortable for chronic ambulatory wear, and prone to electrode impedance degradation as conductive gels desiccate. To enable discrete integration into a comfortable wearable headband—analogous to consumer-grade form factors such as the Muse headset, which demonstrated 76.12% accuracy and 86.66% recall in live streaming validation—EpiPredict incorporates two primary cranial sensing innovations: two-stage channel-aware Set Transformer montage pruning and bimodal neuro-hemodynamic coupling.

Rather than processing all 18 to 22 scalp channels uniformly, the two-stage channel-aware Set Transformer evaluates inter-channel mutual information and spectral discriminative power per patient, systematically pruning redundant electrodes. On the CHB-MIT benchmark, this channel-aware selection reduces the active sensor count from 18 electrodes down to a mean of **2.8 channels** per patient while maintaining an

80.1% mean sensitivity and a false alarm rate of 0.11 FPR/h under strict division protocols. Simultaneously, because scalp EEG is highly vulnerable to muscular (EMG) and motion artifacts and lacks spatial specificity due to volume conduction across the skull, EpiPredict integrates lightweight functional near-infrared spectroscopy (fNIRS) optodes into the same headband. The fNIRS modality measures localized cerebral hemodynamic responses, specifically capturing preictal surges in oxyhemoglobin (HbO_2) and deoxyhemoglobin (HbR) driven by neurovascular coupling prior to generalized tonic-clonic discharges. While standalone convolutional neural networks applied to fNIRS achieve 96.9% to 100% seizure detection accuracy in controlled settings, empirical fusion of EEG and fNIRS via LSTM architectures yields an average accuracy of **98.3%**, outperforming unimodal EEG (97.6%) and unimodal fNIRS (97.0%), whereas spectral-featured Transformer fusion further amplifies preictal discriminability. When high-resolution pre-surgical localization of drug-resistant seizure onset zones is clinically indicated to initialize spatial priors for the wearable montage, the platform can ingest baseline structural and metabolic neuroimaging, including fluorodeoxyglucose positron emission tomography (FDG-PET), translocator protein PET (TSPO-PET), magnetoencephalography (MEG), and blood oxygen level-dependent functional magnetic resonance imaging (BOLD fMRI).

Peripheral Autonomic Biosensing and Contextual Big Data Fusion

A pivotal analytical finding in contemporary predictive neuroscience is the functional divergence between modalities that excel at *ictal detection* versus those that drive *long-horizon preictal prediction*. Because preictal cortical synchronization frequently triggers early autonomic nervous system (ANS) dysregulation via projections from the insular cortex and amygdala to hypothalamic and brainstem cardiorespiratory centers, peripheral wearable sensors worn on a smart wristband capture systemic precursors long before macroscopic scalp EEG spikes emerge.

The EpiPredict peripheral sensor suite integrates six non-invasive physiological modalities interfaced via low-power microcontroller buses such as the STM32F4 or ESP32-S3. Cardiac dynamics are monitored continuously through electrocardiography (ECG) and optical photoplethysmography (PPG), which capture beat-to-beat heart rate (HR), circadian-adjusted residual heart rate, pulse wave transit time, and time/frequency-domain heart rate variability (HRV) metrics reflecting preictal vagal withdrawal and sympathetic dominance. Sympathetic sudomotor arousal is quantified via electrodermal activity (EDA) sensors that measure eccrine sweat gland conductance, decomposed mathematically into slow-varying **tonic skin conductance level (EDAt)** and rapid **phasic skin conductance responses (EDAp)**. Multimodal EEG-EDA coupling directly mitigates EEG motion artifacts while encoding preictal autonomic arousal. Furthermore, three-axis accelerometry (ACC), gyroscope (GY), and surface electromyography (EMG) sensors quantify subtle myoclonic tremors, motor restlessness, and tonic muscle rigidity while providing reference motion vectors for adaptive noise cancellation in cranial EEG and optical PPG channels. Finally, peripheral

infrared skin temperature (TEMP) sensors track thermoregulatory shifts and peripheral vasoconstriction linked to hypothalamic circadian and preictal modulation.

Crucially, explainable artificial intelligence (XAI) evaluations utilizing SHapley Additive exPlanations (SHAP) across longitudinal multimodal cohorts—comprising 25 patients and 40 million data samples predicting seizures 30 minutes to 4 hours in advance—reveal a profound inversion of feature importance between acute detection and multi-hour forecasting. While cranial EEG and cardiac ECG dominate acute ictal detection (achieving 98.5% detection accuracy), they account for only **20%** of the variance in 30-minute to 4-hour preictal prediction. Peripheral HR and HRV contribute **10%** to predictive accuracy, whereas **contextual big data telemetry**—comprising sleep architecture staging, anti-seizure medication non-adherence logs, nutritional and food intake, physical over-exertion, and self-reported or autonomically inferred anxiety states—dominates predictive performance with a **70%** contribution. Incorporating this 70/20/10 multimodal variance distribution enables an 83.0% long-horizon prediction accuracy (with 60% seizure-class recall and 99.0% interictal specificity), further boosted by stratifying patient-specific diurnal (awake) versus nocturnal seizure phenotypes.

Big Data Chronobiology: Circadian and Multiday Phase Locking

Beyond short-term preictal windows spanning minutes to hours, longitudinal big data recordings acquired over months to years demonstrate that epileptic seizures do not occur at random Poisson intervals; rather, they are tightly governed by endogenous chronobiological rhythms. Continuous multimodal recordings spanning a mean duration of 232 days per subject (capturing an average of 76 electrographic seizures per patient via implanted Responsive Neurostimulation [RNS] ground-truth detectors paired with commercial wrist wearables) reveal pronounced **circadian (~24-hour)**, **circa-weekly (~7-day)**, **bi/tri-weekly (14–21-day)**, and **monthly (~28–30-day)** physiological cycles.

Because endogenous physiological cycles drift relative to clock time and fluctuate around a central period tendency in the absence of an external *Zeitgeber*, EpiPredict incorporates wavelet time-frequency decomposition and bandpass filtering centered at each candidate period T_c with a ± 25% bandwidth tolerance ([0.75 T_c, 1.25 T_c]). The instantaneous phase $\phi(t)$ of each brain and wearable recording channel is extracted via the Hilbert-transform analytic signal z(t):

$$z(t) = x_{\text{bp}}(t) + j\mathcal{H}\{x_{\text{bp}}(t)\} = A(t)e^{j\phi(t)}$$

where $x_{\text{bp}}(t)$ is the composite bandpass-filtered physiological timeseries, $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform, A(t) is the instantaneous amplitude envelope, and $\phi(t) \in [-\pi, \pi)$ represents the instantaneous cycle phase. Using circular statistics, the degree of seizure phase locking across N_s recorded seizures occurring at phases $\{\phi_1, \phi_2, \ldots, \phi_{N_s}\}$ is quantified by the resultant vector amplitude (R-value):

$$R = \left| \frac{1}{N_s} \sum_{k=1}^{N_s} e^{j\phi_k} \right|, \quad R \in [0,1]$$

Linear regression modeling between brain-wearable magnitude-squared coherence $C_{xy}(f) = \frac{|P_{xy}(f)|^2}{P_{xx}(f)P_{yy}(f)}$ and the phase-locking resultant vector R confirms that peripheral autonomic cycles covary with intracranial electrographic activity (IEA). Across clinical subjects, physiological signals rank in a strict six-tier hierarchy for high-risk versus low-risk seizure discrimination and subject inclusion: **Tier 1 is Intracranial Electrographic Activity (IEA)**, followed by **Tier 2 Residual Heart Rate (Residual HR)**, **Tier 3 Peripheral Skin Temperature (TEMP)**, **Tier 4 Heart Rate (HR) and Phasic EDA (EDAp)**, **Tier 5 Accelerometry (ACC)** (which exhibits strong discrimination in a smaller subset of subjects), and **Tier 6 Tonic EDA (EDAt) and Heart Rate Variability (HRV)**. Crucially, instantaneous seizure probability reaches its absolute maximum during periods of constructive phase co-occurrence—when an individual simultaneously enters the preferred high-risk phase of their circadian cycle and the high-risk phase of their multiday (e.g., circa-weekly or monthly) cycle. Integrating this slow-varying chronobiological prior with fast-varying EEG and autonomic deep learning outputs enables stratified, personalized forecasting across both diurnal and nocturnal seizure phenotypes.

Signal Pre-Processing and Mathematical Feature Engineering

Noise Suppression and Spectral Band Decomposition

Raw electrophysiological signals acquired in ambulatory environments are heavily contaminated by baseline drift, ocular blink artifacts (0.5 – 4 Hz), frontal and temporal myogenic EMG interference (>30 Hz), and ambient electromagnetic power-line noise (50Hz in Europe/Asia; 60Hz in North America). Within the EpiPredict preprocessing stage, multi-channel EEG signals s(t) are modeled mathematically as a superposition of N non-stationary neural oscillatory components corrupted by additive noise n(t):

$$s(t) = \sum_{i=1}^{N} a_i(t) \sin\left(2\pi f_i t + \phi_i(t)\right) + n(t)$$

where $a_i(t)$, f_i, and $\phi_i(t)$ denote the instantaneous amplitude, frequency, and phase of the i-th cortical oscillator. To isolate epileptiform preictal signatures while preserving high-frequency gamma oscillations, the continuous stream—standardized at a sampling rate of f_s = 256Hz with 16-bit resolution—undergoes zero-mean DC component removal (as DC offsets bear no relation to ictogenesis) followed by fourth-order Butterworth bandpass filtering between 0.5Hz and 100Hz. Power-line interference and its first harmonic are eliminated via cascaded second-order infinite impulse response

(IIR) band-stop notch filters configured with transfer function $H(s) = \prod \frac{s^2+\omega_0^2}{s^2+(\omega_0/Q)s+\omega_0^2}$, excising 47 -- 53 Hz and 97 -- 103 Hz in 50 Hz regions, or 57 -- 63 Hz and 117 – 123 Hz in 60Hz regions.

Absolute, Relative, and Ratiometric Power Spectral Density Formulations

Following temporal segmentation into 5-second to 30-second sliding windows (for example, 5s × 256Hz = 1,280 samples per channel, forming a two-dimensional spatial-temporal matrix $X \in R^{C\times 1280}$), time-domain signals are transformed into the time-frequency domain via the Short-Time Fourier Transform (STFT) or Welch's Power Spectral Density (PSD) estimator across K = 8 sub-bands spanning the five canonical physiological rhythms: **Delta (δ, 0.5 -- 4 Hz)**, **Theta (θ , 4 --}8 Hz)**, **Alpha (α, 8 -- 13 Hz)**, **Beta (β, 13 -- 30 Hz)**, and multi-band **Gamma (γ, 30 -- 100 Hz)**.

To capture subtle cross-frequency energy redistribution characteristic of preictal cortical recruitment, three families of logarithmic spectral features are extracted per electrode channel using the MNE and custom Python signal processing pipelines. First, the **Absolute Spectral Power (P_{absolute})** is computed as the natural logarithm of the summed PSD coefficients within each frequency band $i \in \{1, 2, \dots, 8\}$:

$$P_{\text{absolute}}(i) = \log\left(\sum_{\omega\in\text{band } i} \text{PSD}(\omega)\right), \quad i = 1,2,\dots,8$$

Second, the **Relative Spectral Power (P_{relative})** normalizes the energy in band i against the total broadband spectral energy across all bands on a logarithmic scale, suppressing inter-subject skull conductivity and electrode impedance variance:

$$P_{\text{relative}}(i) = \log\left(\frac{\sum_{\omega\in\text{band } i}\text{PSD}(\omega)}{\sum_{\omega\in\text{all bands}}\text{PSD}(\omega)}\right), \quad i = 1,2,\dots,8$$

Third, **Inter-Band Spectral Power Ratios ($R_{i,j}$)** quantify the pairwise logarithmic power differential between every combination of bands i and j, yielding $\binom{8}{2} = 28$ pairwise combinations:

$$R_{i,j} = P_{\text{absolute}}(i) - P_{\text{absolute}}(j), \quad \forall i < j \in \{1,\dots,8\}$$

Together, these formulations yield exactly 8 + 8 + 28 = 44 highly compact, scale-invariant spectral features per electrode channel, which can be augmented with bivariate cross-channel synchronization features (resulting in a 735-dimensional feature space in high-density montages) or fed directly as STFT spectrograms into two-dimensional convolutional extractors.

Mitigating Severe Class Imbalance and Data Leakage

A pervasive methodological flaw in early seizure prediction literature is improper handling of the extreme class imbalance between interictal recordings (which span

hundreds of hours) and preictal intervals (which last only 20 to 30 minutes per seizure). Naively applying random k-fold cross-validation across overlapping sliding windows introduces catastrophic **temporal data leakage**, wherein adjacent overlapping windows from the same preictal event are split across training and test sets, artificially inflating reported sensitivity.

To guarantee prospective validity, EpiPredict enforces strict **seizure-independent chronological partitioning** and **lead-seizure clustering**. Under the inter-seizure clustering rule, if the temporal interval between two consecutive seizures is less than 1 hour (< 60min), the subsequent seizure is not treated as an independent event but is merged with the preceding lead seizure cluster, preventing post-ictal contamination of the preictal training buffer. Furthermore, asymmetric variable-overlap sampling is restricted exclusively to the *training phase*: preictal intervals of short duration are oversampled using a 5-second sliding window with variable step overlap to generate an equinumerous 1:1 balance against randomly sampled interictal segments, preserving true data distribution characteristics while exposing the model to sufficient preictal patterns. Crucially, during the *validation and test phases*, overlapping segments are strictly prohibited, and evaluation proceeds chronologically on continuous, non-overlapping windows. To further regularize deep representations against limited preictal sample sizes, contrastive self-supervised learning (CLResNet), generative diffusion models for synthetic preictal spectrogram synthesis, domain-adversarial transfer learning, and class-weighted cross-entropy loss functions are integrated into the training pipeline.

Deep Learning and Neuromorphic Algorithmic Design

Hybrid Spatio-Temporal Deep Architectures (CNN-LSTM and Transformers)

Traditional machine learning classifiers—such as Support Vector Machines (configured with a linear kernel and regularization parameter tuned over {0.01, 0.1, 1, 10} to an optimum of C = 1), Random Forests (n = 107 estimators), and eXtreme Gradient Boosting (XGBoost)—offer rapid training times and strong tabular interpretability on engineered features. However, they struggle to capture high-order non-linear spatio-temporal dependencies directly from noisy multichannel biosignals. Conversely, standalone Convolutional Neural Networks (1D, 2D, and 3D CNNs) excel at extracting localized spatial-spectral spike morphologies from STFT matrices (16 × 1280), but underperform on long-range sequential time-series dynamics. Similarly, while Bidirectional LSTMs (Bi-LSTM) capture both forward and backward temporal context (reaching 99.37% sensitivity and 99.6% specificity in offline evaluations), their bidirectional dependency requires future window lookahead, rendering them computationally inefficient and unsuitable for causal real-time streaming.

To achieve optimal causal real-time inference, the high-capacity cloud and edge-hub tier of EpiPredict utilizes a **Hybrid Temporal 2D-CNN + Unidirectional LSTM / Multi-Head Self-Attention Transformer** architecture implemented in PyTorch and TensorFlow/Keras. In the spatial-spectral convolutional front-end, multi-layer 2D depthwise-separable convolutions with batch normalization, Exponential Linear Unit (ELU) activations, and Convolutional Block Attention Modules (CBAM) extract cross-channel spatial topographies and localized gamma/theta bursts from 5-second STFT tensors. Deep Residual Shrinkage Networks (DRSN) apply soft-thresholding inside residual blocks to dynamically attenuate channel-specific motion artifacts, whereas stacked convolutional autoencoders (CAEs) provide unsupervised dimensionality reduction. Subsequently, in the sequential recurrent and attention back-end, the flattened convolutional feature maps are passed sequentially into a unidirectional Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) network—trained with a batch size of 32, an initial maximum of 100 epochs, early stopping patience of 10 to 15 epochs, and dropout regularization—or into a Transformer encoder utilizing scaled dot-product multi-head self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

This hybrid topology captures both microsecond epileptiform transients via convolutional kernels and multi-minute preictal synchronization build-up via recurrent gating and self-attention memory.

Neuromorphic Spiking Neural Networks (SNNs) and Spiking ConvLSTM (SPCLU)

While hybrid CNN-LSTM and Transformer architectures achieve exceptional predictive sensitivity on workstation GPUs, their deployment on wearable or implantable edge hardware is severely bottlenecked by dense Multiply-and-Accumulate (MAC) operations and continuous static memory access. Conventional deep neural networks process every time sample regardless of signal quiescence, consuming watts of power. Furthermore, post-hoc conversion of pre-trained DNNs into Spiking Neural Networks (SNNs) suffers from three structural flaws: SNN conversion represents sequential temporal data as static 2D images, creating vulnerability to future data leakage; the parent non-spiking DNN imposes a hard upper bound on accuracy; and backpropagation-trained static weights preclude online on-chip patient adaptation.

To overcome these barriers, the edge tier of EpiPredict implements a native, **from-scratch Neuromorphic Deep Spiking Neural Network (dSNN)** built around a three-layer **Spiking ConvLSTM Unit (SPCLU)** followed by two Leaky Integrate-and-Fire (LIF) dense classification layers.

Spike Encoding and Membrane Dynamics

Continuous preprocessed EEG and autonomic inputs are converted into sparse binary spike trains via either a Gaussian random discrete rate encoder or a mixed-signal **Asynchronous Delta Modulator (ADM)** circuit that emits UP and DOWN address-event representation (AER) pulses only when the analog waveform changes by a preset delta threshold. Within the SPCLU layers (configured with kernel size 3 × 3 and progressive filter depths $f_1 = 16$, $f_2 = 32$, $f_3 = 64$), spatial-temporal gating is governed by convolutional recurrent operations ($\star$ denotes 2D convolution, $\circ$ denotes the Hadamard product):

$$i_t = \sigma(W_{ii} \star x_t + b_{ii} + W_{hi} \star u_{t-1} + b_{hi})$$

$$f_t = \sigma\big(W_{if} \star x_t + b_{if} + W_{hf} \star u_{t-1} + b_{hf}\big)$$

$$g_t = \tanh\big(W_{ig} \star x_t + b_{ig} + W_{hg} \star u_{t-1} + b_{hg}\big)$$

$$o_t = \sigma(W_{io} \star x_t + b_{io} + W_{ho} \star u_{t-1} + b_{ho})$$

$$c_t = f_t \circ c_{t-1} + i_t \circ g_t, \quad u_t = o_t \circ \tanh(c_t)$$

where u_t acts directly as the somatic membrane potential of the spiking ConvLSTM unit. In the subsequent LIF dense layers, continuous-time sub-threshold membrane dynamics follow the differential equation:

$$\tau_m \frac{dV_i^l(t)}{dt} = -\big(V_i^l(t) - V_{\text{rest}}\big) + R \sum_j w_{ij}^l S_j^{l-1}(t)$$

where $\tau_m = 20$ ms is the membrane time constant, $V_{\text{rest}} = -60$ mV is the resting potential, R is membrane resistance, and w_{ij}^l represents synaptic efficacy. Discretized via the forward Euler method at an empirically optimal timestep resolution of $\Delta t = 16$ ms (yielding a membrane decay factor $\beta = e^{-\Delta t/\tau_m} \approx 0.45$), the discrete LIF state update with hard threshold reset is expressed as:

$$u_t = \beta u_{t-1} + W_{ij} x_{t-1} - z_{t-1} u_{\text{thr}}$$

$$z_t = \Theta(u_t - u_{thr}) = \begin{cases} 1, & u_t \geq u_{\text{thr}} \\ 0, & otherwise \end{cases}$$

To prevent pathological runaway burst firing during high-amplitude artifacts, an **adaptive threshold homeostasis mechanism** dynamically elevates the firing threshold $V_{\text{th}}(t)$ (baseline $V_{\text{th0}} = -50$ mV) following each emitted spike at time t_k:

$$V_{\text{th}}(t) = V_{\text{th0}} + \alpha \sum_{t_k < t} e^{-(t-t_k)/\tau_{\text{th}}}$$

Surrogate Gradient Backpropagation Through Time (BPTT) and Sparsity Regularization

Because the Heaviside step function $\Theta(V(t) - V_{\text{th}})$ is non-differentiable (its Dirac delta derivative is zero everywhere except at the threshold, where it is infinite), standard error backpropagation fails. During the backward pass of truncated Backpropagation Through Time (BPTT, configured over a 20-timestep horizon corresponding to 320ms at $\Delta t = 16$ ms resolution), the discontinuous spike derivative is substituted with a smooth **fast-sigmoid surrogate gradient**:

$$\frac{\partial S(t)}{\partial V(t)} \approx \frac{1}{\left(1 + \beta_{\text{sg}} |V(t) - V_{\text{th}}|\right)^2}$$

This surrogate gradient enables end-to-end optimization using the AdamW optimizer with cosine annealing learning rate decay $\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})(1 + \cos(\pi t / t_{max}))$ while strictly preserving binary event-driven spike propagation during forward inference. Furthermore, because neuromorphic hardware energy consumption scales linearly with spike activity ($E_{\text{total}} \propto N_{\text{neurons}} \cdot N_{\text{timesteps}} \cdot E_{\text{op}}$), the training objective combines class-weighted classification loss (or membrane potential mean-square error against u_{thr}) with an L2 **firing-rate sparsity regularizer** that penalizes deviations from a sparse target firing rate $r_{\text{target}} = 10$ Hz ($\lambda = 0.01$):

$$\mathcal{L}_{total} = -\sum_{c} w_c y_c \log(p_c) + \lambda \sum_{l,i} \left(\frac{1}{T} \sum_{t=1}^{T} z_{l,i}(t) - r_{\text{target}} \right)^2$$

AURA On-Device Continuous Learning and Closed-Loop Calibration

A major impediment to universal seizure prediction is that preictal signatures vary drastically across individuals and evolve over time due to medication adjustments, glial scarring, or chronobiological drift. Transmitting raw 24/7 multichannel EEG to cloud servers for periodic model retraining violates patient privacy and quickly exhausts wearable battery reserves.

To achieve autonomous personalization directly at the edge, EpiPredict incorporates the **AURA (Adaptive, Unlabeled, and Real-time Approximate-Learning)** paradigm paired with a closed-loop false-positive calibration filter. Within the teacher-student on-chip loop, because automated *seizure detection* (identifying an active ictal event during or immediately after onset) is significantly easier and more reliable than prospective *prediction* (reaching 98.5% accuracy), an embedded high-specificity detection model acts as an algorithmic "human expert" to generate training labels on the fly. As physiological data stream through a local circular memory buffer (loop recorder),

whenever the detector confirms an ictal onset at time t_0, it retroactively labels the buffered segment [t_0 - (SPH + SOP), t_0 - SPH] as a true **preictal** training instance.

These on-the-fly auto-generated labels are paired with loop-recorded biosignals to fine-tune the synaptic weights of the plastic SNN output layer directly on-chip using local surrogate-gradient or artificial meta-plastic learning rules. Empirical validation on mixed-signal neuromorphic circuits demonstrates that cross-patient adaptation requires calibrating only the output layer parameters on a **single recorded seizure** from the new patient. Furthermore, correlating past predictive alerts against confirmed detector outcomes in a self-corrective feedback loop with risk-controlling prediction calibration reduces false-positive alarm rates by **15% to 20%**.

Experimental Evaluation, Benchmark Datasets, and Clinical Metrics

Benchmark Big Data Repositories

To ensure rigorous, reproducible validation across diverse patient demographics, recording modalities (scalp versus intracranial), and clinical centers, the EpiPredict framework is benchmarked against five foundational neurological big data repositories, summarized in Table 2.

Table 2: Characteristics of Big Data Neurological and Multimodal Wearable Repositories Utilized for Training, Cross-Validation, and Prospective Benchmarking

Dataset Repository	Cohort Size & Demographics	Recording Modality & Channels	Sampling Rate & Resolution	Total Recording Volume & Seizures	Primary Role in EpiPredict Validation
CHB-MIT Scalp EEG Database	23 pediatric and young adult patients (aged 1.5–22 years) with	Non-invasive scalp EEG; 18 to 22 common channels (10–20 system)	256 Hz, 16-bit resolution	844 to 980+ hours continuous EEG; 163 annotated seizures	Primary benchmark for SPH/SOP sensitivity, FPR/h, and 2.8-channel Set Transformer reduction

Dataset Repository	Cohort Size & Demographics	Recording Modality & Channels	Sampling Rate & Resolution	Total Recording Volume & Seizures	Primary Role in EpiPredict Validation
	intractable seizures				
EPILEPSIAE European Database	37 to 275+ patients across multi-center European EMUs	Long-term scalp sEEG and invasive intracranial iEEG + ECG	250–2,500 Hz (decimated to 256 Hz)	>5,120 hours of continuous multi-day recordings; >209 seizures	Multi-day chronobiological validation, transfer learning, and AURA edge adaptation
Siena Scalp EEG Database	Adult cohort with focal and generalized epilepsy	Non-invasive scalp EEG; 29 channels (including 17 frontal and temporal channels)	512 Hz (downsampled to 256 Hz)	128 hours of recording; 47 annotated clinical seizures	Cross-dataset generalization testing for LSTM and hybrid spatio-temporal networks
TUH EEG Seizure Corpus (TUSZ)	Large-scale heterogeneous clinical hospital	Clinical multi-montage scalp EEG (TCP and	250–512 Hz	Thousands of clinical sessions across diverse seizure	Pre-training foundation backbone and evaluating SNN performanc

Dataset Repository	Cohort Size & Demographics	Recording Modality & Channels	Sampling Rate & Resolution	Total Recording Volume & Seizures	Primary Role in EpiPredict Validation
	population	referential)		morphologies	e across 6 seizure patterns
Multimodal Wearable & Chronobiology Cohorts	25-patient local healthcare cohort + 10-subject ultra-long-term RNS/wearable cohort	EEG, ECG, EMG, EDA (tonic/phasic), PPG (HR/HRV), ACC, TEMP + sleep/adherence logs	Multi-rate biosignal sampling	40 million multimodal samples; mean 232 days/subject (mean 76 seizures/subject)	SHAP feature attribution (30 min–4 hr horizon) and circadian/multiday phase-locking analysis

Formalization of SPH, SOP, and k-of-n Post-Processing

A critical methodological distinction highlighted in recent scoping reviews is the frequent conflation of segment-based classification metrics with event-level clinical forecasting metrics. Evaluating a model solely on window-by-window accuracy or zero-horizon (SPH = 0) classification fails to reflect clinical reality. In rigorous prospective evaluation—standardized by Maiwald et al. and adopted throughout EpiPredict—the continuous physiological timeline progresses sequentially from the **Interictal Baseline** (spanning continuous hours or days) into the **Preictal Transition**, where a predictive alarm is triggered at time t_{alarm}. Subsequent evaluation is governed by two strict temporal intervals following t_{alarm}: the **Seizure Prediction Horizon (SPH)** and the **Seizure Occurrence Period (SOP)**.

The **Seizure Prediction Horizon (SPH)**, also termed the *intervention lead time*, is the mandatory interval [t_{alarm}, t_{alarm} + SPH] (typically configured in the range of 3 to 10

minutes) between the issuance of a predictive warning and the beginning of the expected seizure window. During the SPH, the patient takes protective behavioral action, administers fast-acting rescue medication, or the closed-loop neurostimulator delivers pre-ictal suppression pulses. Immediately following the SPH is the **Seizure Occurrence Period (SOP)**, defined as the interval [t_{alarm} + SPH, t_{alarm} + SPH + SOP] (typically configured between 20 and 30 minutes, though tested across 10 to 55 minutes in cross-patient studies) during which the seizure is expected to initiate. For a prediction to be scored as a **True Positive (TP)** event, the actual electrographic seizure onset t_{onset} must occur strictly *after* the SPH and *within* the SOP:

$$t_{\text{alarm}} + \text{SPH} \leq t_{\text{onset}} \leq t_{\text{alarm}} + \text{SPH} + \text{SOP}$$

If an alarm is triggered and no seizure occurs within the SOP window, the alarm is scored as a **False Positive (FP)**. Conversely, if a seizure occurs without a preceding alarm whose SOP covers t_{onset}, it is scored as a **False Negative (FN)**.

Selecting the durations of SPH and SOP involves a fundamental neuro-ergonomic trade-off between intervention readiness and psychological distress. If the SPH is too short (e.g., < 2min), the patient lacks sufficient time to pull over a vehicle or reach a safe environment. Conversely, if the SOP is excessively wide (e.g., 60 -- 120 min), the patient is subjected to prolonged psychological distress and immobilizing anxiety waiting for a seizure that may occur at any moment over a two-hour span. While clinical trials frequently evaluate discrete pairs such as **Trial 1 (SPH = 5 min, SOP = 25 min)**, **Trial 2 (SPH = 10 min, SOP = 20 min)**, or **Standard Configuration (SPH = 3 min, SOP = 30 min)**, mathematical optimization of the SPH/SOP trade-off by Alaei et al. demonstrates that the global optimum across patient cohorts occurs at **SPH = 7 minutes** and **SOP = 27 minutes**. To prevent isolated noisy 5-second windows from triggering spurious alarms, EpiPredict applies the **k-of-n post-processing voting filter**: an official patient alarm is raised only when at least k out of n consecutive sliding-window predictions exceed the preictal probability threshold within the evaluation buffer, outperforming legacy cumulative firing power post-processing (which yielded only 49% sensitivity and 2.12 FPR/h on EPILEPSIAE).

Event-level **Sensitivity (Sen)** and the **False Prediction Rate per Hour (FPR/h)** are then defined strictly over the uncontaminated interictal recording duration $L_{interictal}$ (in hours):

$$\text{Sensitivity (Sen)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\%, \quad \text{FPR/h} = \frac{\text{FP}}{L_{\text{interictal}}}$$

Comparative Performance Analysis Across State-of-the-Art Architectures

Evaluating the progression of deep learning and neuromorphic models across the CHB-MIT, Siena Scalp EEG, and EPILEPSIAE datasets defines an **Optimal Clinical Utility Zone** characterized by event sensitivity ≥ 90% and false alarm rates ≤ 0.15 FPR/h.

Whereas baseline convolutional architectures and uncalibrated ensembles exhibit higher false alarm rates—such as 0.82, 0.67, and 0.80 FPR/h for uncalibrated baseline CNN, CNN+LSTM, and CNN+CBAM models in early cross-patient experiments, or 75.0% accuracy for 30-second sliding-window CNN-GRU models and 87.6% accuracy for Convolutional Autoencoder + LSTM pipelines—modern hybrid spatio-temporal and attention-guided architectures achieve exceptional clinical specificity. Similarly, while general patient-independent ensemble models (n = 107 estimators) achieve 82.07% sensitivity with 0.0799 FPR/h at a 5-minute SPH and 30-minute SOP, personalizing the ensemble to patient-specific preictal patterns elevates sensitivity to 89.31% while compressing false alarms to 0.03 FPR/h. On the adult Siena Scalp EEG dataset, standalone LSTM networks reach 81.03% accuracy, 72.72% precision, 76.19% recall, and a 74.42% F1-score, whereas advanced spatio-temporal deep networks reach 99.67% accuracy. Table 3 synthesizes the comparative empirical performance of machine learning, deep learning, multimodal, and neuromorphic architectures across benchmark datasets, demonstrating how the EpiPredict architecture surpasses its foundational 70% accuracy threshold.

Table 3: Comprehensive Benchmark Comparison of Seizure Prediction Architectures Across Datasets, Temporal Horizons, and Hardware Efficiency Regimes

Model Architecture & Study	Dataset & Input Modality	SPH / SOP Configuration	Sensitivity / Recall (%)	False Prediction Rate (FPR/h)	Accuracy / AUC / Specificity	Key Architectural & Clinical Distinction
Foundational EpiPredict Target (Samuelson G)	Multimodal Wearable EEG & Biosensors	Real-time continuous risk scoring	≥ 70.0% target	Minimized via edge filtering	≥ 70.0% Acc; 50% event reduction	Baseline clinical translation benchmark for ambulatory management
STFT + 2D CNN	CHB-MIT &	Standard SPH /	89.80 %	0.170 /h	High cross-	Pioneering time-

Model Architecture & Study	Dataset & Input Modality	SPH / SOP Configuration	Sensitivity / Recall (%)	False Prediction Rate (FPR/h)	Accuracy / AUC / Specificity	Key Architectural & Clinical Distinction
(Truong et al.)	Freiburg (Scalp/iEEG)	30 min SOP			dataset stability	frequency spectrogram 2D convolution baseline
1D Binary CNN (BSDCNN)	CHB-MIT (Scalp EEG)	Standard preictal window	88.89 %	0.390 /h	Low memory footprint	Binary single-dimensional weights reduce memory but elevate FPR/h
Patient-Specific Ensemble (n=107 estimators)	EPILEPSIAE / Multichannel EEG	5 min SPH / 30 min SOP	89.31 % (82.07 % general)	0.030 /h (0.0799 /h general)	High patient-specific stability	Demonstrates 2.6 × FPR reduction via patient-specific retraining
Transformer-Guided CNN	CHB-MIT (18-Ch STFT	Standard SPH/SOP	91.50 % (Yan et al.:	0.145 /h	93.50 % AUC	Combines local CNN receptive fields

Model Architecture & Study	Dataset & Input Modality	SPH / SOP Configuration	Sensitivity / Recall (%)	False Prediction Rate (FPR/h)	Accuracy / AUC / Specificity	Key Architectural & Clinical Distinction
	Scalp EEG)		92.11 %)			with global Transformer self-attention
Compact Multi-Head Attention	CHB-MIT (Event-level Scalp EEG)	20 min SPH / 30 min SOP	94.10 % – 94.78 %	0.050 /h	95.80 % Accuracy	Lightweight attention heads achieve ultra-low false alarm rate
Deep Residual Shrinkage Net + GRU	CHB-MIT (Scalp EEG)	10 min SPH / 30 min SOP	91.50 %	0.060 /h (Basri & Arif)	89.70 % Spec; 94.3% Acc	Soft-thresholding shrinkage blocks suppress ambulatory EEG noise
CNN + SVM & SVM+CNN+LSTM Hybrids	CHB-MIT (Scalp EEG)	21 min – 34 min prediction time	92.70 %	Low event FPR	90.8% Spec; 97.07 % Acc	Combines deep spatial-temporal extraction with margin

Model Architecture & Study	Dataset & Input Modality	SPH / SOP Configuration	Sensitivity / Recall (%)	False Prediction Rate (FPR/h)	Accuracy / AUC / Specificity	Key Architectural & Clinical Distinction
						classification
Temporal CNN + LSTM	CHB-MIT (Scalp EEG)	3 min SPH / 30 min SOP	97.75 % – 98.21 %	0.237 /h	Outperforms standalone CNN/ LSTM	Temporal CNN enriches spatial features prior to sequential LSTM gating
Spatio-Temporal Hybrid DL	CHB-MIT & Siena Scalp EEG	5 min SPH / 30 min SOP	98.52 % – 99.31 %	0.010 – 0.045 /h	99.54 % AUC; 98.78 %– 99.70 % Acc	Upper-bound cloud/workstation performance across pediatric and adult cohorts
Two-Stage Set Transformer	CHB-MIT (**2.8 Channels** selected)	3 min SPH / 30 min SOP	80.10 % (Mean; 79.3% All)	0.110 /h	84.4% electrode reduction (18 → 2.8)	Enables discrete, ergonomic headband wearables with minimal

Model Architecture & Study	Dataset & Input Modality	SPH / SOP Configuration	Sensitivity / Recall (%)	False Prediction Rate (FPR/h)	Accuracy / AUC / Specificity	Key Architectural & Clinical Distinction
						channel count
Multimodal XGBoost + LSTM + Feedback	25 Patients (EEG, ECG, EMG, EDA, HR, HRV + Context)	30 min – 4 hr Advance Horizon	83.0% Pred / 98.5% Det	Reduced 15%–20% via loop	99.0% Non-seizure Specificity	Demonstrates 70% SHAP dominance of contextual factors in multi-hour forecasting
Bimodal EEG + fNIRS LSTM (Sipal et al.)	Multimodal Neuro-Hemodynamic	Preictal oxygenation + electrical	>98.0 %	Minimal	98.30 % Accuracy	Surpasses unimodal EEG (97.6%) and fNIRS (97.0%) via neurovascular fusion
Neuromorphic	CHB-MIT (Gaussian	Standard	95.10 %	~0.08 /h	99.20 % Spec;	**98.58% reduction** in computat

Model Architecture & Study	Dataset & Input Modality	SPH / SOP Configuration	Sensitivity / Recall (%)	False Prediction Rate (FPR/h)	Accuracy / AUC / Specificity	Key Architectural & Clinical Distinction
Spiking-CNN	Spike-Encoded EEG)	preictal horizon			0.912 AUC	ional complexity vs. standard CNN

Edge Deployment, Neuromorphic Hardware, and TinyML Optimization

Model Compression: Quantization, Pruning, and Kolmogorov-Arnold Networks

Deploying deep predictive models onto battery-powered wearable microcontrollers (MCUs) or implantable pulse generators requires bridging a four-order-of-magnitude gap in compute and thermal budgets—transitioning from cloud GPUs (such as an NVIDIA Volta operating at 250 W and costing $9,000) down to embedded ultra-low-power silicon operating below 50 mW and costing between $3 and $15. Within the EpiPredict C++ and TinyML deployment workflow, trained floating-point (FP32) networks undergo a three-stage compression pipeline utilizing TensorFlow Lite (TFLite Micro), QuantLab, and DORY (Deployment Oriented to MemorY) compilers.

First, structural magnitude pruning excises redundant synaptic connections and low-salience convolutional filters via Alternating Direction Method of Multipliers (ADMM) optimization and activity regularization, shrinking parameter counts to between 0.41M (for SDCAE-Bi-LSTM) and 0.72M. Second, post-training and quantization-aware fixed-point conversion transforms weights and activations from 32-bit floating-point to 8-bit integer (INT8) or binary/integer representations, or substitutes multiplications with energy-efficient Adder Networks. Empirical evaluation of the quantized **EPIDENET** architecture across CHB-MIT and PEDESITE datasets (using 8-second windows of 2,048 time samples at 256 Hz trained with the SSWCE loss function) confirms that INT8 quantization via QuantLab and TFLite induces a negligible accuracy shift of only **±0.1%** in sensitivity and specificity while reducing task energy consumption by over 90%. Similarly, embedded C++ XGBoost implementations on ESP32-S3 wearables maintain numerical equivalence to Python floating-point inference within 1.19×10^{-7} in probability output, achieving 67% event sensitivity (12/18 focal-

to-bilateral tonic-clonic seizures) with a false-alarm rate of 0.31 per 24 h and a 61.5 s median detection latency across 2,643.69 hours of validation recording. Third, for extreme memory-constrained microcontrollers operating under a few-shot personalization regime (K = 128), replacing traditional multi-layer perceptrons with spline-parameterized Kolmogorov-Arnold Networks (KAN) compresses the total model footprint to just **11.76 kB** and executes inference in **4.6 ms** on a 32-bit MCU, achieving 94.87 ± 0.31% accuracy, 94.90 ± 0.31% F1-score, and an AUC-ROC of 0.904 ± 0.005.

Hardware Benchmarking Across Edge MCUs, RISC-V Clusters, and Neuromorphic Silicon

A critical engineering finding emerges when benchmarking quantized seizure prediction networks across heterogeneous edge architectures: standard single-core microcontrollers (such as the ARM Cortex-M4F or Cortex-M7) suffer from prolonged active execution times during multichannel convolution, which inflates static leakage energy. Conversely, parallel ultra-low-power RISC-V clusters and event-driven mixed-signal neuromorphic processors complete inference in a fraction of the time or operate asynchronously in the sub-milliwatt domain, as detailed in Table 4.

Table 4: Empirical Hardware Benchmarking of TinyML and Neuromorphic Processors for Real-Time Edge Seizure Prediction

Hardware Platform & Processor Core	Target Architecture / Model	Active Power Dissipation	Inference Latency (Time / Window)	Energy per Inference	Throughput & Efficiency Metrics	Clinical Form Factor Suitability
Green Waves GAP9 (1 + 9 × RISCY RISC-V Cluster)	Quantized EPIDENET (INT8 via QuantLab + DORY)	**17.89 mW** average	**2.84 ms**	**0.051 mJ**	**726.45 MMAC/s**; **40.61 GMAC/s/W** (160 × vs. ARM)	Ideal for continuous 24/7 wearable EEG headband & smart glasses

Hardware Platform & Processor Core	Target Architecture / Model	Active Power Dissipation	Inference Latency (Time / Window)	Energy per Inference	Throughput & Efficiency Metrics	Clinical Form Factor Suitability
Green Waves GAP8 (1 + 8 ×CV32E40P RISC-V)	Quantized EPIDENET (INT8 via QuantLab + DORY)	~38 mW	6.82 ms	0.260 mJ	305.83 MMAC/s; 7.86 GMAC/s/W	High-efficiency wearable wristband / neckband hub
32-Bit Embedded MCU with Tiny KAN	Compressed Few-Shot KAN (K=128, **11.76 kB** size)	< 20 mW	**4.60 ms**	< 0.10 mJ	94.87 % Acc; 0.904 AUC-ROC	Ultra-compact patch, smart tattoo, or behind-the-ear sensor
STM32F4 / ARM Cortex-M4F	TFLite Micro Quantized CNN / EPIDENET	50 – 150 mW (0.05--0.15 W)	100.0 – 405.0 ms	0.15 – 8.09 mJ	1.50 – 4.40 MMAC/s; 0.25 GMAC/s/W	Low-cost ($3–$15) peripheral sensor aggregator (ACC/GY/TEMP)

Hardware Platform & Processor Core	Target Architecture / Model	Active Power Dissipation	Inference Latency (Time / Window)	Energy per Inference	Throughput & Efficiency Metrics	Clinical Form Factor Suitability
ARM Cortex-M7 Single-Core MCU	TFLite Micro Quantized EPIDENET	~250 mW	31.99 ms	8.090 mJ	10.78 MMAC/s; 0.25 GMAC/s/W	Standard embedded medical telemetry node
Mixed-Signal Neuromorphic (DYNAP-SE2 / Custom CMOS SNN)	Event-Driven ADM + Spiking ConvLSTM / 3-Layer SNN	**Sub-milliwatt** (< 1 mW)	**< 10.0 ms** (at $\Delta t = 16$ ms)	**2,400 × lower** than ConvLSTM	80.7% – 98.58 % compute reduction; 64.8% area reduction	Implantable closed-loop neurostimulators (RNS/VNS) & always-on silicon
Raspberry Pi 4 / NVIDIA Jetson Nano (Edge Hub)	TFLite Multimodal Fusion (EEG+PPG+EDA+ACC)	1.5 W – 5.0 W	**2.185 ms** (0.002185 s)	~3.3 -- 10 mJ	Full multimodal fusion + Firebase cloud sync	Bedside home hub or portable belt-worn edge gateway

Executing the quantized EPIDENET network on the **GreenWaves GAP9** processor (utilizing custom DORY memory tiling across its 1 + 9×RISCY multi-core cluster) achieves an inference time of **2.84ms** and consumes just **0.051mJ** per inference at an average power draw of **17.89mW**, representing a **160× improvement in energy efficiency (40.61GMAC/s/W)** over optimal ARM Cortex deployments. Furthermore, when transitioning to mixed-signal neuromorphic processors such as **DYNAP-SE2**, analog waveforms are encoded directly into UP/DOWN pulse trains via Asynchronous Delta Modulators and filtered by spike-based resonant-and-fire neuron blocks (comprising excitatory R-E and inhibitory R-I neurons with slow excitatory synapses) that are selectively sensitive to the 200 -- 600 Hz firing rate regime, extracting partial cortical synchronization (*Chimera states*) in real time even under Gaussian jitter noise ($\sigma = 0.005$ s to 0.015 s). Although deep SNNs exhibit an approximate 8.0% raw accuracy trade-off relative to unconstrained floating-point ConvLSTM networks on complex 6-pattern intracranial datasets, they deliver a **>2,400× reduction in energy consumption**, an **80.7% to 98.58% reduction in computational complexity**, an **over 80% reduction in energy**, and a **64.8% reduction in silicon area occupation**, establishing the definitive hardware pathway for sub-milliwatt implantable closed-loop electroceuticals.

To further conserve flash storage and wireless transmission bandwidth on ambulatory wearables, EpiPredict operates in a **Dual-Mode Selective Telemetry** regime. During normal interictal states, the edge processor discards raw high-rate waveforms after local inference, logging only compressed chronobiological summary statistics; only when the preictal risk score crosses the k-of-n warning threshold does the device activate high-resolution selective data collection, synchronizing the preictal-to-ictal biosignal trace to a secure cloud database (such as Firebase) for neurologist review and continuous model refinement.

Technical Challenges, Mitigation Strategies, and Translational Roadmap

Addressing Core Translational Challenges

Translating predictive neuroengineering platforms from retrospective benchmarks into routine clinical practice requires overcoming four systemic challenges identified in the EpiPredict framework and broader clinical literature. The first challenge is **complex neural signal interpretation and ambulatory artifact contamination**. In free-living environments, masticatory muscle contractions, talking, ocular blinks, and locomotion generate high-amplitude artifacts in scalp EEG, while optical PPG and skin-surface EDA suffer from motion-induced probe detachment and ambient thermal shifts. EpiPredict mitigates this contamination by combining Deep Residual Shrinkage Networks (DRSN) and CNN-based automated EEG artifact removal with cross-modal sensor validation—requiring concurrent preictal signatures across cranial EEG/fNIRS

and peripheral autonomic (EDAp and Residual HR) channels before incrementing the k-of-n alarm counter.

The second challenge involves **pronounced inter- and intra-patient variability**. Preictal duration, focal onset topography, and optimal SOP lengths vary substantially across patients (with optimal SOPs ranging from 10 to 55 minutes and exhibiting high standard deviation across cohorts). Moreover, nocturnal seizures exhibit markedly different autonomic trajectories compared to awake seizures. The platform addresses this variability by stratifying models by vigilance state (nocturnal versus awake), weighting baseline susceptibility by individual circadian and multiday phase-locking vectors, and utilizing the AURA on-device learning loop to calibrate output thresholds on a single patient-specific seizure.

The third challenge centers on **high computational and battery constraints**. Continuous multi-sensor streaming rapidly depletes wearable batteries if processed via unoptimized floating-point networks. EpiPredict resolves this bottleneck through channel-aware Set Transformers that prune cranial electrodes by 84.4% (down to 2.8 channels), paired with INT8 quantization on GAP9 RISC-V processors and surrogate-gradient Spiking-CNNs that reduce compute overhead by up to 98.58%. Finally, to overcome **algorithmic opacity and clinical "black-box" resistance**, EpiPredict integrates SHAP (SHapley Additive exPlanations) values and attention-weight topographic mapping directly into the clinician interface, explicitly quantifying whether a given preictal alert was driven by frontal-temporal gamma bursts, phasic EDA surges, circadian phase co-occurrence, or recent medication non-adherence.

Phased Clinical and Engineering Development Roadmap

Following the developmental trajectory established by Samuelson G, clinical translation of EpiPredict proceeds across four sequential phases. **Phase 1 (Prototype Algorithm and Neuromorphic Simulation)** encompasses the finalization of the hybrid 2D-CNN/Transformer cloud backbone and surrogate-gradient Spiking ConvLSTM (snnTorch / TFLite Micro) edge pipelines, validated against CHB-MIT, Siena, and EPILEPSIAE repositories under strict non-overlapping chronological splits (SPH = 7 min, SOP = 27 min). **Phase 2 (Heterogeneous Multimodal Dataset Expansion)** initiates prospective collection of synchronized 3-channel dry-EEG/fNIRS headband and wrist PPG/EDA/TEMP/ACC telemetry paired with electronic sleep and medication adherence diaries across diverse demographic and underserved regional cohorts to overcome single-center dataset homogeneity. **Phase 3 (Institutional EMU Collaborations and Hardware-in-the-Loop Validation)** deploys the GAP9 and STM32F4/ESP32-S3 wearable prototypes alongside gold-standard inpatient video-EEG and implanted RNS systems across partner comprehensive epilepsy centers to verify prospective false-alarm rates (< 0.05 FPR/h) and single-seizure AURA calibration. Finally, **Phase 4 (Continuous Closed-Loop Refinement and Regulatory Translation)** integrates homomorphic encryption and federated learning protocols for privacy-preserving cross-institutional model updates, culminating in prospective

randomized clinical trials evaluating quality-of-life improvements and unexpected seizure reduction (≥ 50%) under FDA Software as a Medical Device (SaMD) and EQUATOR Network reporting standards.

Ethical Governance, Author Declarations, and Compliance Statements

Ethical Considerations and Algorithmic Governance

The deployment of continuous neuro-surveillance and predictive analytics in vulnerable neurological populations necessitates stringent ethical safeguards across biometric privacy, informed consent, and demographic fairness. Raw electroencephalographic and autonomic biosignals constitute highly sensitive biometric identifiers capable of revealing cognitive states, emotional arousal, and psychiatric comorbidities. By executing real-time inference and AURA adaptation directly on-device via TinyML and neuromorphic processors, EpiPredict preserves edge data sovereignty and eliminates the need to stream continuous raw EEG to external cloud servers. When selective cloud synchronization is utilized for clinician review or federated aggregation, transmissions are protected via homomorphic encryption and zero-trust differential privacy.

Simultaneously, because false-positive predictive alerts can induce acute anticipatory anxiety, panic, or unnecessary rescue medication ingestion, dynamic informed consent protocols must transparently communicate patient-specific Positive Predictive Values (PPV) and FPR/h metrics. The mobile application and web interface allow patients and neurologists to customize alert sensitivity thresholds—for example, shifting between high-sensitivity mode during hazardous activities and ultra-low-false-alarm mode during rest. Furthermore, because historical seizure prediction models trained on small, homogeneous cohorts (such as 15 to 25 patients from a single locality) risk degraded sensitivity across underrepresented age groups, skin phototypes (which affect optical PPG and fNIRS signal-to-noise ratios), and focal versus generalized seizure etiologies, EpiPredict mandates multi-center validation across pediatric (CHB-MIT) and adult (Siena, EPILEPSIAE, TUH) cohorts alongside domain adaptation to ensure equitable diagnostic performance.

Author Declarations

Author Attribution and Contributions: SAMUELSON G conceived the EpiPredict platform architecture, formulated the multi-layered neurological predictive analytics workflow, defined the clinical and quantitative performance targets, designed the integration of multimodal wearable sensing with deep learning and neuromorphic edge computing, and authored the manuscript.

Ethics Approval and Consent to Participate: All benchmark analyses reported in this study utilize publicly accessible, de-identified human neurological repositories

(including the CHB-MIT Scalp EEG Database, the Siena Scalp EEG Database, and the Temple University Hospital [TUH] EEG Seizure Corpus) or established institutional datasets (EPILEPSIAE) that were collected under prior Institutional Review Board (IRB) approvals and informed patient consent protocols in accordance with the Declaration of Helsinki. All methodological reporting adheres to relevant EQUATOR Network guidelines for biomedical artificial intelligence research.

Data and Code Availability Statement: The open-access benchmark datasets analyzed in this framework are publicly available via PhysioNet (CHB-MIT and Siena Scalp EEG) and the Temple University Hospital EEG resources portal (https://isip.piconepress.com/projects/tuh_eeg/), while the European EPILEPSIAE database is accessible to licensed research institutions via https://www.epilepsy-database.eu/. Reference implementations for Spiking ConvLSTM units (SPCLU) are accessible via the open-source snnTorch Python package, alongside TensorFlow Lite Micro and DORY deployment toolchains for RISC-V and ARM Cortex microcontrollers.

Competing Interests and Funding Declaration: The author (**SAMUELSON G**) declares no competing financial or non-financial interests that could inappropriately influence the design, analysis, or reporting of the EpiPredict platform.

Conclusion

The **EpiPredict: Advanced Neurological Predictive Analytics Platform**, developed by Samuelson G, establishes a rigorous, clinically actionable blueprint for transitioning epileptic seizure forecasting from retrospective inpatient analysis to autonomous, real-time ambulatory care. By synthesizing sparse, channel-aware cranial sensing (2.8-channel EEG coupled with hemodynamic fNIRS) and peripheral autonomic wearables (PPG, EDA, ECG, TEMP, ACC) with longitudinal big data chronobiology, the architecture captures both millisecond cortico-electrical precursors and multi-day physiological phase-locking cycles that govern ictal susceptibility. Crucially, the recognition that contextual and autonomic telemetry drive 70% to 80% of multi-hour predictive variance—whereas cranial EEG dominates acute short-horizon detection—resolves a long-standing limitation of unimodal EEG forecasting. Finally, by pairing high-sensitivity hybrid CNN-LSTM and Transformer extractors (achieving up to 99.31% sensitivity and 0.01FPR/h) with surrogate-gradient Spiking ConvLSTM networks and quantized RISC-V TinyML processors (delivering 2.84ms inference latency at 0.051mJ and a 2,400× neuromorphic energy advantage), EpiPredict surmounts the power-latency wall of wearable edge computing. Coupled with AURA on-chip continuous adaptation and closed-loop false-alarm calibration, this interdisciplinary synthesis of predictive neuroscience, biomedical big data, and neuromorphic engineering provides a scalable pathway to exceed 70% ambulatory

forecasting accuracy, halve unexpected epileptic events, and restore safety and autonomy to millions of individuals living with refractory epilepsy.

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